CDOT TDM Calculator

User Guide

A step-by-step guide to estimating vehicle-miles-traveled (VMT) reduction from transportation demand management strategies for projects in Colorado.

Prepared for the Colorado Department of Transportation, Office of Innovative Mobility.

Access the application using the link provided by CDOT.

1. Before you start

What the calculator does

The CDOT TDM Calculator estimates how much a package of transportation demand management (TDM) strategies would reduce driving — vehicle miles traveled, or VMT — for a specific project area in Colorado. It combines Colorado-specific baseline data with published, peer-reviewed effect sizes so that the resulting estimates are source-backed and can be cited.

It is built to support PD 1601 analyses, Multimodal Transportation and Mitigation Options Fund (MMOF) and similar grant applications, corridor and community planning, and early conversations about which strategies are worth pursuing.

What it is — and what it is not

The calculator is a **screening and comparison tool**. It produces defensible planning-level estimates that let you compare strategies and build a credible package. It is not a substitute for a travel demand model run, a corridor study, or design-grade analysis, and its outputs are not survey-grade measurements of what a project will achieve.

Estimates are most reliable when used to compare options and to establish a reasonable order of magnitude. Treat a single headline number as an informed planning estimate, not a guarantee.

What you will need before you begin

The calculator fills in everything it can from state and federal data as soon as you select your project area — baseline VMT, commute mode share, population and employment density, average trip length, and transit service levels. You do not need to look any of that up.

What it cannot know is the **specifics of the project you are proposing**. Most strategies ask for one or two inputs that describe your project, and those usually have to come from outside the tool. Gathering them before you start will make the process much faster.

Common examples:

- **Bicycle and pedestrian strategies** — the share of roadway in your study area that would receive bike lanes, sharrows or a protected facility, and how much of the area's traffic uses those streets.
- **Transit strategies** — the location, span and frequency of the routes you plan to improve, the share of the area's routes affected, and any fare or pass-subsidy details.
- **Parking strategies** — your proposed pricing schedule and the boundary of the area where it applies.

- **Employer and programmatic strategies** — the size of the workforce or population your program will reach, and the value of any incentive offered.
- **Land use strategies** — the housing or job counts in your development program, against local baseline density.

You do not have to find these on your own. Every input has a **How to find this value** panel explaining what the number means and where to look, and several link directly to the relevant CDOT resource — the Traffic Data Explorer and Highways: Traffic Counts for roadway volumes, and the Statewide Transit Routes and Transit Points maps for transit service. Where a reasonable published default exists, the field is pre-filled with it.

If you cannot find a precise number

Use a defensible estimate and write down how you arrived at it in the Source / justification field beside the input. A documented assumption is far stronger in a grant application than an undocumented precise-looking figure.

Getting access

The calculator runs in a web browser — use the link provided by CDOT. There is nothing to install and no account to create. It works on desktop, tablet and mobile; on a phone the map appears above the panel and the strategy filters collapse into a “Filter strategies” control.

2. How the tool is organized

The calculator follows three steps, shown as numbered tabs across the top of every screen:

- **1 Area selection** — choose the zones your project covers.
- **2 Strategy selection** — browse strategies, configure them, and add them to your package.
- **3 Results** — review the combined effect and export it.

A metrics bar beside the tabs always shows your project area, baseline VMT, the package's combined impact, and the annual reduction, so you can see the effect of any change immediately.

You can move between the steps freely. Returning to the map to adjust your area does not discard the strategies you have configured. **Start over**, at the top right, clears everything and asks for confirmation first.

Your work is not saved automatically

The calculator does not store projects between visits, and there is no login. Export a PDF report or CSV before you close the tab if you want a record.

3. Step 1 — Define your project area

Everything the calculator estimates is anchored to a project area you define on the map. The area is expressed as a set of traffic analysis zones (TAZs) — the small geographic units the Colorado Statewide Travel Demand Model uses. Each TAZ carries its own travel and demographic data, which is what lets the tool pre-fill your baseline.

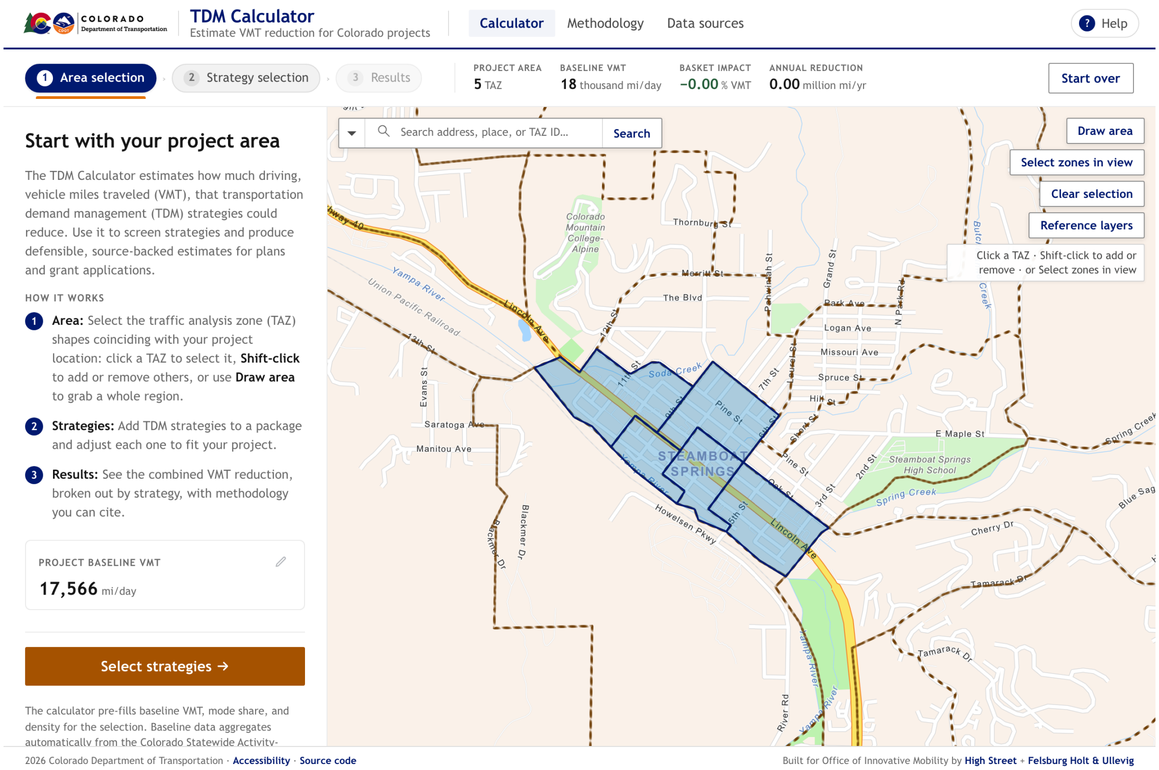

Figure 1. The Area selection step: five zones selected along Lincoln Avenue in Steamboat Springs.

Four ways to select zones

- **Click** a zone to select it. Clicking a different zone replaces the selection.
- **Shift-click** to add a zone to the selection, or to remove one already selected.
- **Draw area** to outline a region: click each corner on the map, then double-click to close the shape. Every zone inside is added.
- **Select zones in view** to add every zone currently visible. Pan and zoom to frame your area first — this is the quickest route for a whole town or corridor, and it needs no mouse precision.

You can also find a location with the **search box**, which accepts a street address, a place name, or a TAZ ID. Searching moves the map; it does not select anything by itself.

Reference layers adds optional context to the map — transit routes and stops, for example — to help you judge which zones belong in your area. These layers are for orientation only and never affect your results.

Choosing the right zones

Select the zones your project actually touches. For a corridor project, that is usually the zones the corridor runs through and those immediately adjacent, where trips would plausibly shift. For a development or campus, it is the zones containing the site.

Be deliberate about how wide you go. Baseline VMT is the denominator for every percentage the tool reports, so **selecting a broader area makes the same strategy look smaller**. A bike lane serving one neighborhood, measured against a whole county's VMT, will appear to do almost nothing. Match the area to the population the project genuinely serves.

What fills in automatically

As soon as you select zones, the calculator aggregates their data and pre-fills your baseline: daily and annual VMT, commute mode share from the U.S. Census, population and employment density, average trip length, average vehicle occupancy, and transit service levels. You will see these values again as Project context on each strategy page.

Overriding baseline VMT

If you have a better figure for your area — from a corridor study or a local model run — you can replace the modeled baseline. Select the pencil beside **Project baseline VMT** in the left panel, enter your value, and record where it came from in the note field. The note is not decorative: it travels into the exported report, and it is what allows a reviewer to accept a number that differs from the state model.

4. Step 2 — Build your strategy package

With an area selected, choose Strategy selection. The catalog contains 26 strategies across seven categories: Transit, Bike and Ped Infrastructure and Amenities, Land Use, Shared Mobility, Supportive and Programmatic, Parking Management, and Roadway Capacity.

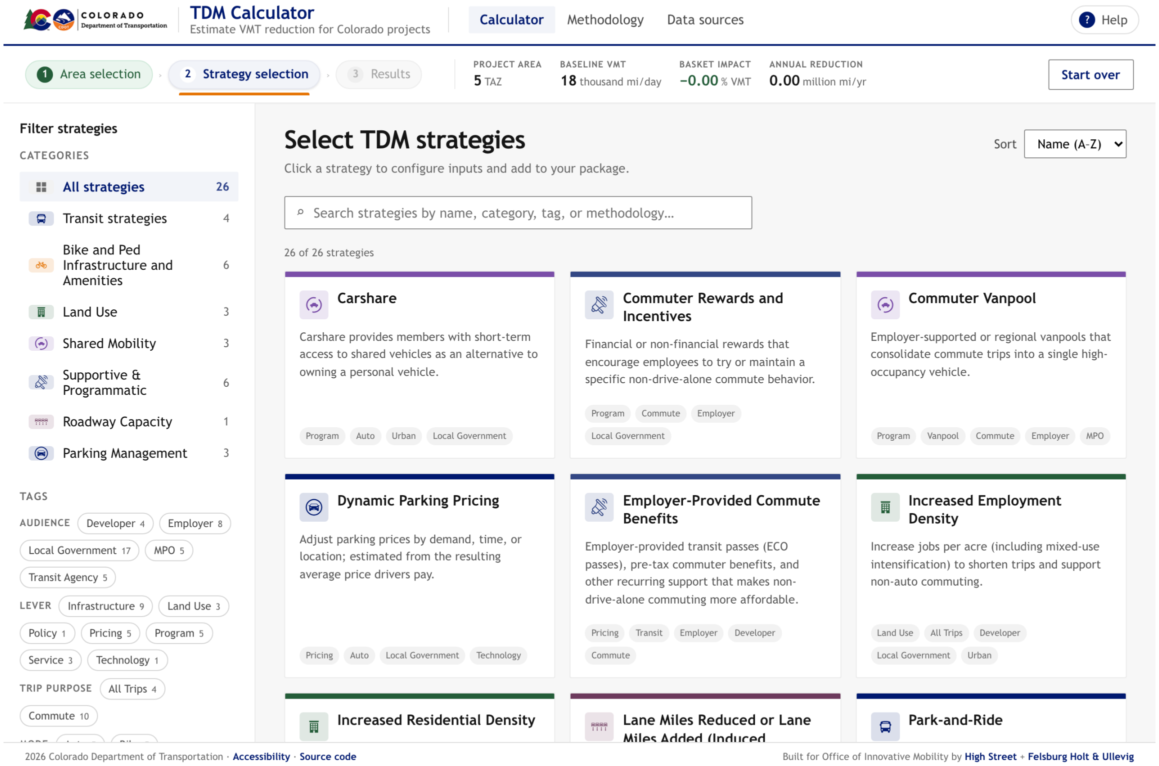

Figure 2. The strategy catalog, filterable by category, tag and free-text search.

Finding the right strategies

Filter by category to narrow the list, or use the tag filters to cut across categories — by who the strategy targets, what lever it pulls, or which trip purposes it affects. The search box matches strategy names, categories, methods and tags. On a phone, these controls sit behind the "Filter strategies" heading; select it to expand them.

Reading a strategy page

Selecting a strategy opens its full page. The layout is consistent:

- **Description and applicability** — what the strategy is, the area types it suits (shown as Recommended context chips), and when to use it instead of a related strategy.
- **Configure** — the inputs that describe your project.
- **Project context** — the baseline values drawn from your selected zones that feed the calculation.
- **At your settings** — the live estimated effect, updating as you change inputs.
- **Methodology** — the method behind the number, its source, and any cap that applies.

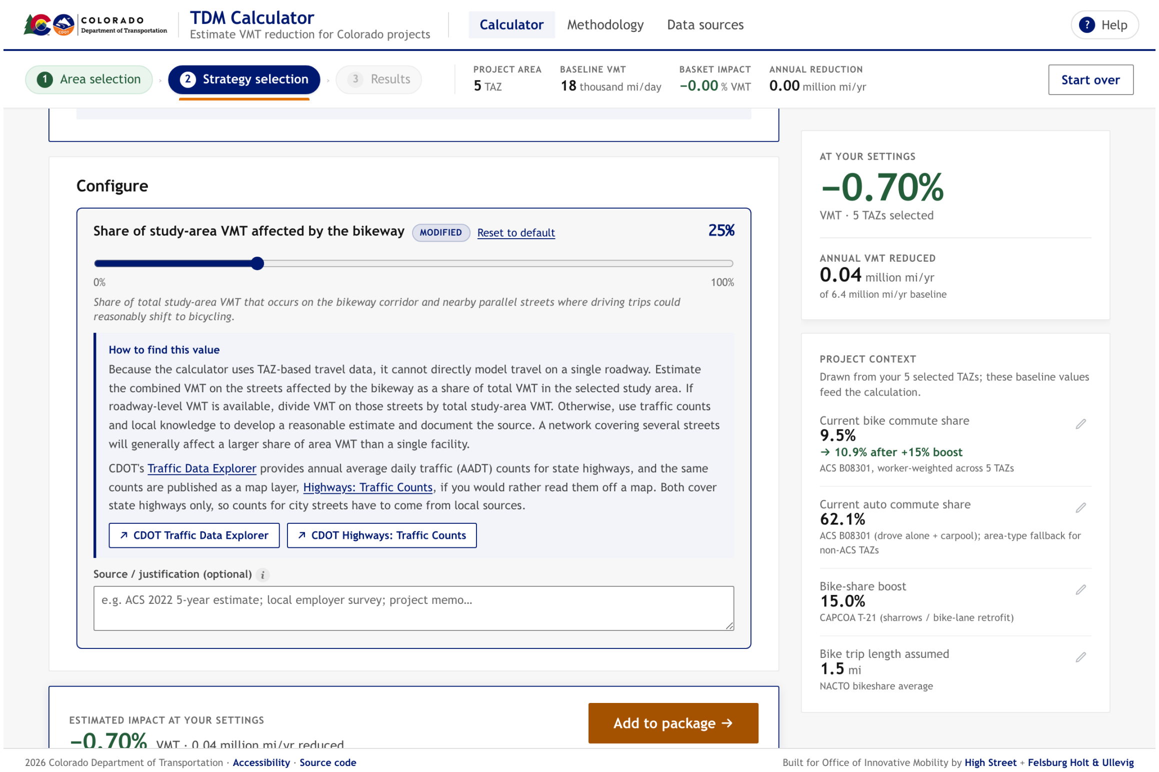

Figure 3. Configuring a strategy: the input, its guidance, and the justification field.

Setting the inputs

Inputs arrive pre-filled with a published default or a value derived from your area. Change one and a MODIFIED badge appears beside it, with a Reset to default link — so at a glance you can always tell your assumptions from the tool's.

Some strategies have a **Prerequisite** input that must be set before the estimate means anything; it is marked as such. The **How to find this value** panel under each input explains the number in plain language and, where one exists, links straight to the CDOT dataset that holds it.

Documenting your assumptions

Beside each input is a **Source / justification** field. It is optional to the software and close to essential in practice.

Anything you enter is carried into the exported PDF report, next to the value it explains. When a reviewer asks why you assumed a quarter of the area's traffic would be served by a bikeway, the answer travels with the estimate rather than living in someone's memory. A short, specific note — the study, the survey, the site plan, the conversation with the transit agency — is enough.

The same applies to **Project context**. If a baseline value does not reflect conditions you know on the ground, select the pencil beside it, enter your figure, and explain why. Overrides apply to that strategy's calculation and are reported alongside it.

Adding to your package

When the inputs look right, select **Add to package**. You are returned to the catalog and the metrics bar updates. Add as many strategies as your project includes; you can revise any of them later from the Results page without losing the rest.

5. Step 3 — Read your results

The Results step reports what your package achieves as a whole.

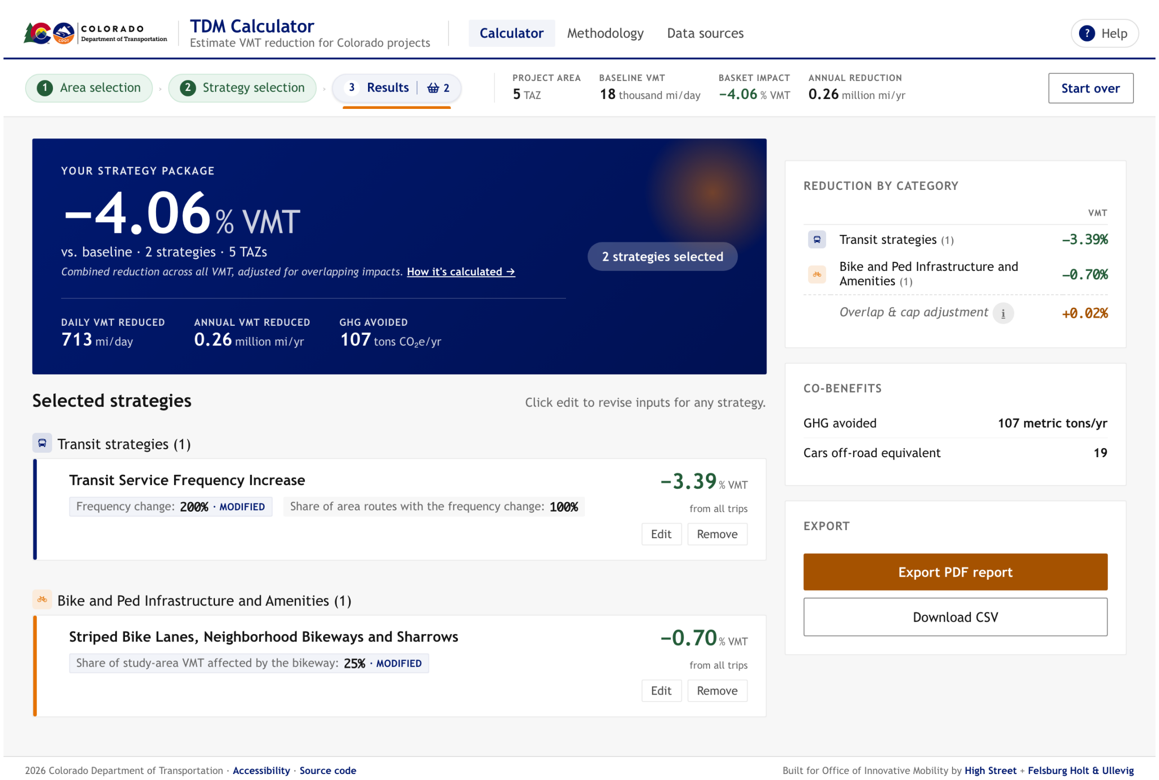

Figure 4. The Results step: headline reduction, per-strategy contributions, and exports.

The headline figures

The large percentage is the combined reduction in VMT across your project area, measured against its baseline. Beneath it are the same result in daily and annual vehicle miles, and the greenhouse gas emissions avoided — converted at 0.412 kg CO_2e per vehicle mile, the blended Colorado rate from EPA MOVES — reported in metric tons per year alongside a cars-off-road equivalent.

Below the headline, each strategy is listed with the inputs you chose and its own contribution, grouped by category. Edit returns you to that strategy's page; Remove drops it from the package.

Why the total is less than the sum of its parts

This is the single most common question a reviewer will raise, and the calculator is explicit about it.

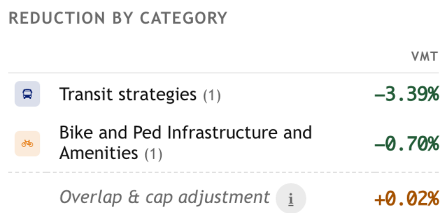

Figure 5. Overlap and cap adjustments are shown as their own line, not hidden in the total.

Two strategies that both move the same trips cannot each claim the full effect. The calculator combines them using CAPCOA methodology, which dampens overlapping effects and applies a maximum to each category — a subsector cap. The difference between the simple sum of your strategies and the combined total appears as its own line, Overlap & cap adjustment, so nothing is quietly absorbed.

Where two strategies act on the same travel in a way worth reviewing, a **Potential overlaps** note appears beneath the strategy list explaining which ones and why. It is guidance, not an error — you may have good reason to keep both.

When the result is an increase

Some strategies increase driving. Added roadway capacity induces demand, and the calculator models that. If your package produces a net increase, the headline turns orange and reads as an increase rather than a reduction. That is a legitimate finding, and reporting it honestly is part of the tool's purpose.

6. Exporting and citing your work

Export PDF report produces a formatted report containing your project area and a map of it, the baseline, every strategy with the inputs and justification notes you entered, the per-strategy and combined results, the co-benefits, and the method and source behind each figure. This is the artifact to attach to a grant application. Your browser's print dialog opens; choose Save as PDF.

Download CSV produces the same results as data, for your own analysis or to combine scenarios in a spreadsheet.

Both exports record the assumptions behind the numbers, which is what makes them defensible later. Two pages in the application provide the deeper reference and can be cited directly:

- **Methodology** — how strategies are calculated, how they are combined, the caps that apply, and the sources behind each method.
- **Data sources** — every dataset behind the baseline, with vintage, provider and what it contributes.

7. Reference

Glossary

VMT	Vehicle miles traveled — total miles driven by motor vehicles. The calculator's primary measure.
TAZ	Traffic analysis zone — the geographic unit of the state travel model, and the unit you select on the map.
Baseline VMT	Daily VMT in your project area before any strategy is applied. The denominator for every percentage reported.
Mode share	The proportion of commute trips made by each mode — driving alone, carpool, transit, bicycle, walking — from the U.S. Census.
AVO	Average vehicle occupancy — average persons per vehicle. Higher occupancy means the same travel produces less VMT.
Area type	A zone's classification as urban core, urban, suburban or rural, derived from population and employment density. Several strategies vary by area type.
Elasticity	How responsive travel behavior is to a change — for example, how much transit ridership rises when fares fall.
Subsector cap	A maximum combined reduction for a category of strategies, preventing implausible totals when several are stacked.
CAPCOA	The California Air Pollution Control Officers Association, whose published handbook provides many of the effect sizes used here, adapted to Colorado conditions.
Induced demand	New driving generated by added roadway capacity — an increase in VMT rather than a reduction.

Where the numbers come from

- **CDOT Statewide Travel Demand Model (2019)** — baseline VMT, trips, trip length, vehicle occupancy, the trip-purpose split, and the roadway network by facility class.
- **U.S. Census, American Community Survey (table B08301)** — commute mode share for each zone.
- **NOAA NCEI 1991–2020 climate normals** — days per year suitable for cycling, interpolated to each zone; used by the bicycle strategies.
- **CDOT public feature services** — highway, transit and urban-area layers used for context and for several strategy inputs.
- **EPA MOVES** — the emissions rate behind the greenhouse gas co-benefit.

Limitations

- **Estimates are planning-level**. They support comparison and screening, not design or final programming.
- **Results depend on your inputs**. A carefully justified input produces a defensible estimate; a guess produces a guess.
- **The baseline is a 2019 model vintage**. A refresh to a newer base year is expected; check the Data sources page for the current vintage.
- **Coverage is Colorado only**, and zone-based — the tool cannot model a single roadway in isolation.
- **Strategies not in the catalog are not estimated**. Four further strategies are documented but not yet calculable.

Accessibility

The calculator is built to WCAG 2.1 and 2.2 Level AA and Section 508 standards, and an independent accessibility audit is underway; formal conformance will be published as a VPAT once that review is complete. The application can be operated entirely by keyboard — including selecting zones, using the “Select zones in view” control rather than clicking the map — and is built for screen-reader use, with per-view page titles and status messages announced as results update.

The PDF report is generated through your browser's print function, so its accessibility depends on the browser you use. If you need the report in a tagged, fully accessible format, or in any other alternative format, request it through the Accessibility link in the footer of every page.

Getting help

Select **Help** in the top-right corner of any screen for a short tutorial video, this guide, and contact details. The Methodology and Data sources pages answer most questions about how a number was produced.